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Research Paper

An ecological perspective on the temporal variation in Pileated Woodpecker (*Dryocopus pileatus*) drumming behavior in Alberta, Canada

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ABSTRACT. As old-growth forest ecosystems become increasingly scarce in North America, the need to accurately and efficiently survey old-growth specialists and keystone species, such as the Pileated Woodpecker (*Dryocopus pileatus*), becomes increasingly important. A common survey method for birds is to detect auditory cues to determine presence. Therefore, it is important to understand temporal patterns in audible cues, which are often linked to a species' breeding phenology, to optimize survey timing. It is well known that the Pileated Woodpecker, as a non-migratory bird, has a breeding season that begins before most migratory passerines arrive at their breeding grounds. However, the timing of the peak Pileated Woodpecker breeding season, and therefore auditory activity, is relatively unknown at the northern extent of its range. We explored the temporal variation of Pileated Woodpecker drumming behavior using passive acoustic monitoring methods at the northern extent of its range in Alberta, Canada. Peaks in auditory cues were near sunrise (06:00) in early April (2 April). Mean daily temperature and day length were the most influential environmental variables that affected the drumming of Pileated Woodpeckers. Drums were more likely to be detected at daily mean temperatures close to zero degrees Celsius and on days where day length was approximately 13 hours long. Based on these findings, we calculated that a minimum of ten one-min long surveys should be conducted during the peak periods of Pileated Woodpecker activity (near sunrise in early April) to ensure accurate presence/absence data for this species in Alberta, Canada. These guidelines can be used for planning future surveys and methods to utilize existing non-optimized surveys to ensure the accuracy of Pileated Woodpecker site occupancy.

Une perspective écologique sur la variation temporelle du comportement de tambourinage du Grand Pic (*Dryocopus pileatus*) en Alberta, Canada

RÉSUMÉ. Les écosystèmes forestiers anciens se raréfiant en Amérique du Nord, il devient de plus en plus important de recenser avec précision et efficacité les spécialistes de forêts anciennes et les espèces-clés, telles que le Grand Pic (*Dryocopus pileatus*). Une des méthodes courantes de relevés d'oiseaux consiste à détecter les sons qu'ils produisent pour déterminer leur présence. Il est donc important de comprendre la tendance temporelle de la production de ces sons, qui est souvent liée à la phénologie de nidification, afin d'optimiser la période de relevés. Il est bien connu que le Grand Pic, en tant qu'oiseau non migrateur, a une saison de reproduction qui commence avant que la plupart des passereaux migrateurs n'arrivent sur les lieux de nidification. Cependant, le point culminant de la saison de reproduction du Grand Pic, et donc la production de sons, est relativement peu connu dans la partie nord de son aire de répartition. Nous avons étudié la variation temporelle du comportement de tambourinage du Grand Pic à l'aide de méthodes passives d'enregistrement à la bordure nord de son aire de répartition en Alberta, au Canada. La quantité de sons produits a atteint son maximum à l'approche du lever du soleil (06:00) au début du mois d'avril (2 avril). La température quotidienne moyenne et la longueur du jour ont été les variables environnementales qui ont eu le plus d'effet sur le tambourinage des pics. Le tambourinage était plus susceptible d'être détecté aux températures quotidiennes moyennes avoisinant zéro degré Celsius et aux jours dont la durée était d'environ 13 heures. Sur la base de ces résultats, nous avons calculé qu'un minimum de dix relevés d'une durée d'une minute devraient être effectués pendant la période de pointe d'activité du Grand Pic (à l'approche du lever du soleil au début du mois d'avril) afin de garantir l'obtention de données de présence/absence précises pour cette espèce en Alberta, au Canada. Les spécialistes peuvent utiliser ces recommandations pour planifier les futurs relevés et corriger l'utilisation de relevés non optimisés existants afin d'assurer l'exactitude de l'occupation des sites par le Grand Pic.

Key Words: *behavior; bioacoustics; detection probability; drumming; Pileated Woodpecker*

INTRODUCTION

Accurate and reliable data collection for wildlife species is essential for designing and implementing effective monitoring. Wildlife species differ in activity levels throughout the day and year, so a key step in designing a monitoring program is assessing when it will most likely be able to detect the target species. Although useful for determining occurrence and trend for hundreds of species, surveys like the Breeding Bird Survey (BBS) are not optimized for some species because the timing of the surveys is not aligned with peak periods of activity, resulting in

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higher false negative rates and presumably less accurate data, trends, and habitat modeling. Thus, determining when a species is most likely to be detected is crucial for minimizing costs, avoiding additional surveys, and maximizing data quality.

Point counts, conducted through human observation and autonomous recording units (ARUs), primarily rely on each bird species to produce an auditory cue to detect a species' presence. The rationale for the timing of the BBS and many other multispecies songbird surveys (late May to early July; Ziolkowski et al. 2023) is that the greatest number of species are audible at that time because it is the peak breeding period for most songbirds (Ziolkowski et al. 2023). However, vocalization is vital to mate selection and territory defense for many species (Kilham 1959, Kroodsma and Byers 1991, Catchpole and Slater 1995); thus, acoustic signals among species vary throughout the breeding cycle (Catchpole and Slater 1995, Helm et al. 2006, Tremain et al. 2008, Avey et al. 2011, Harms and Dinsmore 2014, Upham-Mills et al. 2020). As such, peak auditory activity does not occur simultaneously across all avian species because of variability in breeding cycles (e.g., different time periods for each stage of the breeding cycle and asynchrony in the timing of the breeding cycle; Perrins 1970, Tremain et al. 2008, Harms and Dinsmore 2014, Agostino et al. 2020, Edwards et al. 2022, Hannah et al. 2022). Furthermore, whether auditory behaviors align with breeding activities is not established for all species. This is especially true for non-migratory species that do not have to travel to breeding grounds to begin the breeding season. Species that vocalize infrequently may have a low probability of detection and are often of particular concern because of limited data (Harms and Dinsmore 2014). Furthermore, species with large home ranges may have a lower detection probability at a given location because they simply are not available to produce an acoustic cue during the time of the survey (Johnson 2008, Edwards et al. 2020). If survey timing does not coincide with peak detection times for such species, they may appear rare when, in fact, they are simply less detectable when most surveys are conducted (Johnson 2008, Harms and Dinsmore 2014, Hedley et al. 2020, Hannah et al. 2022, Jeliazkov et al. 2022).

One such species is the Pileated Woodpecker (*Dryocopus pileatus*), a species receiving increased attention in Canada because of recent changes in federal nest protection regulations (Government of Canada 2022). As a primary cavity excavator, the Pileated Woodpecker plays a keystone role in forest ecosystems by creating cavities that are used by species across taxa (Martin and Eadie 1999, Bonar 2000, 2001, Aubry and Raley 2002, Martin et al. 2004, Aitken and Martin 2007). Thus, researchers and land managers have emphasized the importance of monitoring this species more effectively (Aubry and Raley 2002, Aitken and Martin 2007, Trzcinski et al. 2022). Pileated Woodpeckers use auditory cues to attract mates and establish a breeding territory (Kilham 1959, Tremain et al. 2008). The Pileated Woodpecker typically produces two types of auditory cues: calling and drumming (Kilham 1959). Pileated Woodpecker calling has been associated primarily with foraging but is also used to communicate between breeding pairs (Kilham 1959). Drumming is believed to occur primarily during mating displays and territorial defense but may also occur outside of the breeding season (Kilham 1959, Tremain et al. 2008). Foraging and excavation may sound like drumming in that the individual strikes a surface with their beak, but these taps do not produce the distinguishable rhythmic pattern of a drum (Kilham 1959). We focus on the variation of Pileated Woodpecker drumming patterns due to their role in mate selection and territory defense, both important life history traits associated with the breeding cycle. Thus, understanding temporal patterns of drumming may help inform us when and why Pileated Woodpeckers are using particular areas.

Over the past decade, the use of ARUs has increased (Shonfield and Bayne 2017). If deployed over an extended period of time, ARU sampling can occur at a larger temporal scale (both in frequency and length of the sampling periods) to detect species throughout a much larger part of the day and year than traditional point counts. Little research has been done to evaluate how ARUs might be used to monitor Pileated Woodpecker occurrence and behavior. As a non-migratory species, it is well established that Pileated Woodpeckers are heard at many different times of the year, but whether there are distinct peaks in activity is important to establish. Tremain et al. (2008) suggested that Pileated Woodpecker drumming patterns in Florida follow a similar seasonal pattern to that of many passerines, peaking in March. In Oregon, activities associated with breeding (e.g., cavity excavation, mate selection, territory defense) have been observed to begin in March/April (Bull and Meslow 1988). However, because of the extensive latitudinal range of the Pileated Woodpecker, it is necessary to understand if these patterns differ at the northern edge of the species' range where spring conditions are later. Furthermore, it is important to establish if Pileated Woodpeckers demonstrate behavioral patterns like other birds on a daily cycle (i.e., dawn chorus). With an extensive North American range, the Pileated Woodpecker experiences a wide array of environmental conditions, so the timing of peak drumming behavior may vary geographically (Tremain et al. 2008, Bull and Jackson 2020). Thus, evaluating the environmental cues that prompt a species to alter its drumming behavior is important. Whether the Pileated Woodpecker relies on photoperiod (Dawson 2008, Da Silva and Kempenaers 2017, Welklin et al. 2023), average temperature (Bruni et al. 2014, Boelman et al. 2017, de Zwaan et al. 2022), snowpack (Boelman et al. 2017), or seasonal rainfall (Welklin et al. 2023) as cues to begin giving acoustic cues for breeding is unclear. Additionally, it is unknown if Pileated Woodpeckers alter their drumming behavior in response to daily changes in weather conditions (de Zwaan et al. 2022).

We address these knowledge gaps with the following three objectives: (1) Evaluate how Pileated Woodpecker drumming behavior changes on both daily and seasonal scales. To achieve this objective, we evaluated how drumming varies as a function of the hour of the day and day of the year. By doing this, we determined the effort required to detect a Pileated Woodpecker at different times of the day and different days of the year. (2) Identify the environmental variables that best predict variation in Pileated Woodpecker drumming behavior on a seasonal/yearly basis. In doing so, we test if Pileated Woodpecker drumming frequency is best explained by (a) static variables (low interannual variation) like Julian date, day length, and expected mean daily temperature; (b) dynamic variables (strong interannual variation) like mean temperature, snow cover, and days from green-up; or (c) a combination of static and dynamic variables. (3) Explain the degree to which Pileated Woodpecker drumming behavior is affected by daily variations in weather (i.e., precipitation and deviation from the expected normal mean temperature).

METHODS

Study area

Our study area included sites across the western Canadian boreal forest and Rocky Mountains of Alberta, Canada. Upland boreal forest ecosystems in this area are generally dominated by jack pine (*Pinus banksiana*), trembling aspen (*Populus tremuloides*), or white spruce (*Picea glauca*; Larsen 1980, Krebs et al. 2001). Lowland boreal forests (i.e., muskegs, bogs, or fens) are dominated by black spruce (*Picea mariana*) and tamarack (*Larix laricina*; Larsen 1980, Krebs et al. 2001, Chen and Popadiouk 2002). Areas in the Rocky Mountains and foothills are characterized by more elevational variation and include three distinct ecozones: montane, subalpine, and alpine. We only examined locations in the montane ecozone within the Rocky Mountain and foothills regions, however. The montane ecozone is typically made up of upland pine (lodgepole pine; *Pinus contorta*), spruce (white spruce), or mixed wood (containing spruce/pine and trembling aspen) forests (Habeck 1987).

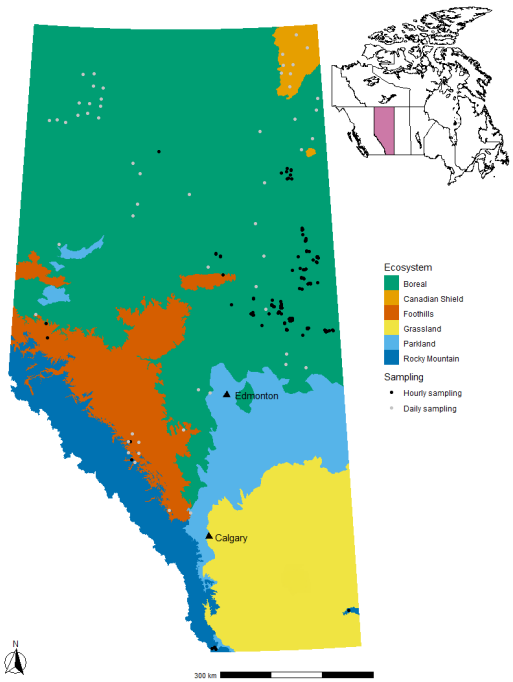
Acoustic data collection and processing

One ARU (model SM2, SM3, & SM4, Wildlife Acoustics Inc., Maynard, Massachusetts, USA) was deployed at each location in Figure 1. These ARUs passively recorded stereo audio from two microphones with 16 dB gain, 24 kHz sampling rate, and 16-bit depth audio settings. Originally deployed for multispecies monitoring conducted by the Alberta Biodiversity Monitoring Institute (ABMI), we randomly selected locations at least 600 m from each other and which previously had detected Pileated Woodpeckers. In other words, we selected locations where Pileated Woodpeckers were known to be present. Each location was sampled during one year between 2015 and 2022.

For sites selected for seasonal analysis, ARUs were deployed in March and retrieved in July. The date of recordings processed for this analysis included Julian dates 60–219 (1 March–7 August). We programmed ARUs to record one-min recordings between the approximate time of sunrise and 2 hours after each day. This time frame was chosen because many multispecies monitoring programs, such as ABMI, collect much of their data during this period. By focusing on recordings within this time frame, we ensured access to numerous recordings and provided more applicable results for these organizations. We sampled these recordings by randomly selecting one recording per site per day.

We deployed ARUs at sites for the analysis of daily patterns in late May/early June and retrieved them within a few days of deployment; thereafter, we deployed that ARU at a different location during this sampling period. This sampling design allowed us to sample more locations with a limited number of ARUs. The date of recordings processed for this analysis included Julian dates 77–191 (18 March–10 July). The units deployed at these locations had a more frequent recording schedule, recording one min every half-hour during the morning (~03:00–10:00) and every other hour during the rest of the day. We took a subsample of these recordings to analyze one to four full days of recordings from each location (mean: 1.5, standard deviation: 0.54). One full day included every recording on the second day of the ARU's deployment (at least 10). Ten recordings was the minimum number of recordings because we believed that it would adequately capture the variation in daily patterns if ARUs were programmed to record audio approximately every hour (10–49 recordings/site, mean 23 recordings/site). Greater than 10 recordings per location were opportunistically processed on the basis of the frequency of recording schedule or deployment length, thus maximizing detections of Pileated Woodpeckers. ARUs used within this analysis typically frequently moved across locations over the length of the breeding season. By doing so we were able to sample a larger range of locations with a limited number of ARUs. A total of 110 locations were used across both analyses. Once samples were chosen, we uploaded them to WildTrax (http://www.wildtrax.ca), where processing occurred.

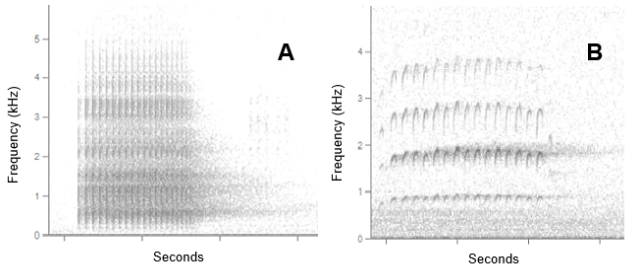
Fig. 1. Map of Alberta, Canada, showing 110 sites where Pileated Woodpeckers were surveyed. Alberta's location within Canada is displayed in purple in the top right corner. Gray dots represent sites (68) where data were collected daily, over a seasonal time frame, for the day-of-year analysis. Black dots represent sites (42) where data were collected hourly for the time-of-day analysis. Broad ecosystems of Alberta are displayed (Government of Canada 2016). Black triangles represent major cities.



We used two types of acoustic processing methods: visual scanning and manual listening. Within Wildtrax, expert observers identified a target species as they processed each recording using visual (via spectrogram) and auditory cues. Observers visually processed each recording's spectrogram to find the spectral signature of Pileated Woodpecker drumming and calling patterns (Fig. 2). An observer could confirm correct identification when auditory cues aligned with the visual identification of the target spectral signatures (Fig. 2). Observers tagged each instance where they could identify a Pileated Woodpecker and labelled whether it was a call or drum (Fig. 2). Three observers processed a total of 8077 one-min recordings. A single observer verified all tagged

vocalizations to ensure consistency and accuracy among identification. Pileated Woodpecker drums are approximately 3 seconds long and consist of 10–30 knocks with consistent cadence and decreasing amplitude (Kilham 1959; Fig. 2). Pileated Woodpecker drums are unlike other woodpecker species within the study area because of the way in which the knocking pattern decreases in amplitude at the end of the drum. Other woodpecker species, such as the American Three-toed Woodpecker (*Picoides dorsalis*) or the Black-backed Woodpecker (*Picoides arcticus*), may appear to have similar drumming patterns, but these species' drums gradually decrease in cadence but not amplitude. Although all types of Pileated Woodpecker auditory cues were identified and tagged, this study focused on the drumming cues because of the low frequency of calling behavior and frequent association with drumming (only 70 recordings with observations of calling, compared to the 249 recordings with observations of drumming). Furthermore, most recordings where a Pileated Woodpecker call was detected, a drum was also detected. This study only used locations that recorded Pileated Woodpeckers presence at least once.

Fig. 2. Example spectrograms of a Pileated Woodpecker drum (A) and "wuk" series call (B) in Alberta, Canada. The spectrograms were generated from WildTrax (https://wildtrax.ca/).



Environmental and ecological predictor variables

We assessed how various ecological and environmental variables influenced Pileated Woodpecker drumming behavior. Specific variables included: hours from sunrise, day length, mean temperature, normal expected mean temperature, snow cover, days from green-up, precipitation, and deviation from expected normal temperature. A full description of these variables and which analyses they were used for is available in Table 1.

We calculated the time since sunrise for each recording with the *suncalc* package in the program R (Thieurmel and Elmarhraoui 2022). Sunrise is when the sun crests the horizon, whereas sunset is when the sun is entirely below the horizon (Thieurmel and Elmarhraoui 2022). Time since sunrise is the difference between sunrise and recording time in hours. Day length was calculated as the time elapsed between sunrise and sunset.

We downloaded daily mean temperature values for each recording from Wildtrax. Wildtrax extracts weather data from the closest publicly available weather station from Environment and Climate Change Canada. In addition, we determined the normal expected mean temperature from the *weathercan* R package (LaZerte and Alberts 2018). We extracted daily expected normal temperatures from the nearest publicly accessible temperature station for each day of the year. The normal expected mean temperature is calculated by the average of the daily mean temperatures observed at that weather station for a particular day of year across the years 1981–2010 (LaZerte and Alberts 2018). We calculated the deviation from the expected normal temperature to determine if the weather on that day was hotter or colder than average for that specific time of the year. To do this, we calculated the difference between the daily mean temperature and the normal expected mean temperature. These weather stations also measured daily precipitation (in mm).

Table 1. Predictor variables that describe the relationship between time and Pileated Woodpecker behavior. Objective indicates the study objective the variable was used to achieve.

Predictor variable	Description	Objective
Julian date	Numeric index for the day of the year. 1 January equates to Julian date 1.	1,2
Hour	Hour of the day (1–24).	1
Hours from sunrise	The number of hours from the time the sun crosses the horizon.	1
Day length	The number of hours the sun position was above the horizon.	2
Mean temperature	Average observed daily temperature gathered from nearby weather station data.	2
Normal expected mean temperature	The expected mean temperature for a given date based on historical temperature data.	2
Snow cover	Estimated average snow cover (% of ground covered within approximately 1 km).	2
Days from green-up	The number of days from the green-up date of that location.	2
Precipitation	Amount of daily precipitation (mm) occurring at the nearest weather station.	3
Deviation from the expected mean temperature	The difference in observed temperature and normal expected temperature.	3

We extracted snow cover values from the global snow cover dataset produced by Hengl (2021). This dataset uses MODIS imagery to create near-daily composites of snow cover at a 250-m resolution. The values derived from this dataset represent the fractional snow cover, or the number of pixels determined to be covered by snow at a one-km resolution. With these data, we determined snow cover (within one km) for the nearest available date for each recording.

We derived the green-up date from Friedl et al. (2019) for each site location. This dataset uses MODIS imagery to calculate the enhanced vegetation index (EVI) on a temporal frequency of approximately every other day (from 2000 to 2020) and a spatial resolution of 500 m. We determined the days since green-up by calculating the difference between the recording date and the date when EVI first crossed 15% of EVI amplitude (i.e., the green-up date). We calculated green-up dates for each location within a 500-m buffer based on the year the location was sampled. The 500-m buffer averages the green-up pixel values within the radius.

Seasonal and daily patterns

We utilized generalized additive mixed effect models (GAMMs) to determine the relationship between temporal variables and Pileated Woodpecker drumming behavior. These models apply

smoothing parameters to relationships, allowing non-linear relationships between predictor and response variables (Pedersen et al. 2019). The response variable Pileated Woodpecker drumming behavior was binomially distributed and was defined by whether a Pileated Woodpecker drum was detected (present) or not (absent) during the one-min survey period. Therefore, multiple detections of Pileated Woodpecker drums within a recording were treated the same as a single detection. We treated location as a random effect because each observation at a specific location was not independent. Survey year would be represented within the models by the random effect of location, because locations were only sampled during a single year. We did not include ARU model within our analysis because the ARU models used within this analysis have proven to have similar detection radii for low frequency calls and drumming (Yip et al. 2017). Using a circular spline transformation, we fit the smoothing terms used for the variables hour since sunrise and hour of the day (Pedersen et al. 2019). We implemented a circular spline for these models because the values at the beginning and end of the distribution were equally close to the value minus one (i.e., hour 23 is equally close to hour 0 as it is to hour 22). We used a thin-plate spline for the Julian date model (Pedersen et al. 2019). These models are further described in Table 2. We predicted the peak detection windows from these models by finding the maximum drumming rate. To observe how latitude influenced these relationships, we created additional models that contained an interaction spline (Pedersen et al. 2019) between the model's main effect variable and latitude. We did not include the interaction in further models if there was no significant relationship between latitude and the corresponding variable. Significant P-values were used to determine if drumming rate had a significant ($P \leq 0.05$) relationship with time of day, time since sunrise, and day of year (objective 1).

Surveys until detection confidence calculation

A key question in monitoring is how much effort is needed to demonstrate whether a species is present or absent from a location. Detection probability (P) is the likelihood of observing an individual during a survey and can be calculated by:

$$\frac{\#\ of\ observations}{survey\ effort} \tag{1}$$

This essentially is the number of one-min recordings where a Pileated Woodpecker was detected divided by the total number of recordings. We determined the detection probability for Pileated Woodpeckers using the data available from peak drumming activity periods established via both the seasonal and daily analyses.

We used Eq. 2 to estimate how many one-min surveys (x) were required to obtain high confidence (90% d = 0.1 and 99% d = 0.01) if a Pileated Woodpecker was not detected at a site that the species would not use that site over the course of the breeding season. We calculated the probability of detection and the number of surveys for the peak detection intervals for the hour of the day, hour from sunrise, and Julian date. Additionally, we conducted these calculations for the standard breeding songbird survey periods in June.

Table 2. Model set 1: seasonal and daily patterns. The model set used to examine how Pileated Woodpecker (PIWO) drumming behavior change on a seasonal and daily scale. All models are GAMMs, and notation is described in the "mgcv" R package (Wood 2017). The notation "bs" indicates the base spline smoothing function used, "cc" indicates a circular spline and "re" indicates a random effect spline. The default thin-plate spline ("tp") was used if no base spline was specified.

Model input	Variable	Pattern described
gam(PIWO drum~ s(Julian Date, bs= "cc")+ s (Location, bs= "re"))	Julian date	Seasonal
gam(PIWO drum~ s(Hour, bs= "cc")+ s (Location, bs= "re"))	Hour	Daily
gam(PIWO drum~ s(Time Since Sunrise, bs= "cc")+ s(Location, bs= "re"))	Time since sunrise	Daily

Sliwinski et al. (2016) estimated the number of surveys required (x) to obtain a given confidence level (d) to detect a Pileated Woodpecker during the breeding season if an individual occupied the site. P represents the probability of detecting a Pileated Woodpecker (detection probability) within the peak detection times.

$$x \geq ln(d) * ln(1 - P) - 1 \tag{2}$$

Variables influencing acoustic behavior

We created six GAMMs (Table 3), containing one predictor variable, as well as the random effect of location. We used these GAMMs to evaluate what environmental variables best predicted variation in Pileated Woodpecker drumming. P-values were used to generate a list of candidate models that suggested a statistical correlation with Pileated Woodpecker drumming. We then used the Akaike information criterion (AIC) score to select the model that best explained the data given these candidate models. From this comparison, we determined the most influential static versus dynamic variables that affected Pileated Woodpecker drumming behavior on seasonal/yearly scales. Based on univariable analysis using P-values, we determined the static variables that were significant predictors of drumming were Julian date, day length, and normal expected mean temperature. Julian date and day length are highly correlated variables and we do not expect these models to differ greatly. As an anthropogenic measure of seasonality, Julian date provides us with a more relatable understanding of how seasonality may affect Pileated Woodpecker drumming behavior. However, day length is the more natural measure of Earth's position around the sun. Therefore, we present both models because they provide a slightly different lens to view the relationship (anthropogenic/natural). The significant dynamic variables were mean temperature, snow cover,

Table 3. Model set 2: ecological mechanisms. The model set used to examine what environmental factors drive Pileated Woodpecker (PIWO) drumming behavior to change over a seasonal scale. All models are GAMMs, and notation is described in the "mgcv" R package (Wood 2017). The notation "bs" indicates the base spline in which the smoothing function should be used, "cc" is a type of circular spline and "re" indicates a random effect spline. A thin-plate spline ("tp") is used if no base spline is specified.

Model input	Variable
gam(PIWO drum~ s(Julian Date, bs= "cc")+ s(Location, bs= "re"))	Julian date
gam(PIWO drum~ s(Day length)+ s(Location, bs= "re"))	Day length
gam(PIWO drum~ s(Mean Temperature)+ s(Location, bs= "re"))	Daily mean temperature
gam(PIWO drum~ s(Normal Temperature)+ s(Location, bs= "re"))	Daily normal expected mean temperature
gam(PIWO drum~ s(Snow Cover)+ s(Location, bs= "re"))	Snow cover
gam(PIWO drum~ s(Days from Green-up)+ s(Location, bs= "re"))	Days from green-up

and days from green-up. These variables are expected to shift yearly based on local conditions. To test the importance of static versus dynamic variables, we determined which static and which dynamic variable best explained variation in drumming behavior via AIC comparisons of the univariable GAMMs (Table 3). Then we compared the AIC scores of multivariable models, which included the best static variable and the best dynamic variable.

The effects of daily variation in weather

To determine the effects of daily variations in weather (objective 3), we created two generalized linear mixed models, including location as a random effect (Table 4). One model contained the predictor variable of deviation from the normal expected mean temperature, whereas the other contained precipitation (Table 4). We evaluated the P-values from this generalized linear mixed model to determine if these weather variables influence Pileated Woodpecker drumming behavior. Additionally, we compared AIC values to a null model to further investigate how these models performed. For this analysis, we used generalized linear models because we were not as concerned with model fit as we were in the previous analyses. Rather, we were more interested in detecting a signal (i.e., if the variables influenced drumming behavior). We fit these models following the binomial distribution.

RESULTS

Throughout all our sampling, we identified a total of 249 one-min acoustic recordings that contained a Pileated Woodpecker drum. A total of 133 of these recordings were identified in the time-of-year sampling across 42 locations. Across the time-of-year sampling design we detected a Pileated Woodpecker between one and 14 times at a site (mean: 3.6, standard deviation: 3.51). In the time-of-day sampling, we identified 116 acoustic recordings with a Pileated Woodpecker drum across 68 locations. Across the time-of-day sampling design we detected a Pileated Woodpecker between one and four times at a site (mean: 0.54, standard deviation 0.87). Locations where Pileated Woodpeckers were not observed in this study were either sites where Pileated

Table 4. Model set 3: daily variation effects. The model set of generalized linear models used to examine how daily variations in weather affect Pileated Woodpecker drumming behavior.

Model input	Variable
$\beta_{\text{Normal Temperature Deviation}} + \varepsilon_{\text{Location}}$	Deviation from normal expected mean temperature
$\beta_{\text{Precipitation}} + \varepsilon_{\text{Location}}$	Precipitation

Woodpeckers had been observed in a previous year or locations outside of our sampling period. We found a significant relationship between the day of the year (Julian date) and Pileated Woodpecker drumming ($p < 0.05$; Fig. 3). Both the hour from sunrise and the hour of the day ($p < 0.05$) significantly affected Pileated Woodpecker drumming behavior ($p < 0.05$; Fig. 4). Peak detection of Pileated Woodpecker drums occurred on 2 April (Julian date 92; Fig. 3) and during hours 06:00 or 1 hour since sunrise (Fig. 4). The probability of detecting a Pileated Woodpecker drum on 2 April was approximately 9.00% (Fig. 3). The probability of detecting a Pileated Woodpecker drum at hour 06:00 was 7.76%. The probability of detecting a Pileated Woodpecker drum 1 hour since sunrise was 7.01% (based on an average day of the year in the data set, June 4). There was no significant effect of an interaction with latitude on Pileated Woodpecker drumming, so we did not include this interaction in the models described below.

With these estimated probabilities, we calculated the number of one-min surveys required to achieve 90% and 99% confidence of Pileated Woodpecker presence/absence at a site. During the peak

Fig. 3. The predicted probability of detecting a Pileated Woodpecker drum on a seasonal scale. Predictions generated from the generalized additive model are described in Table 2. The shaded area represents the standard error.

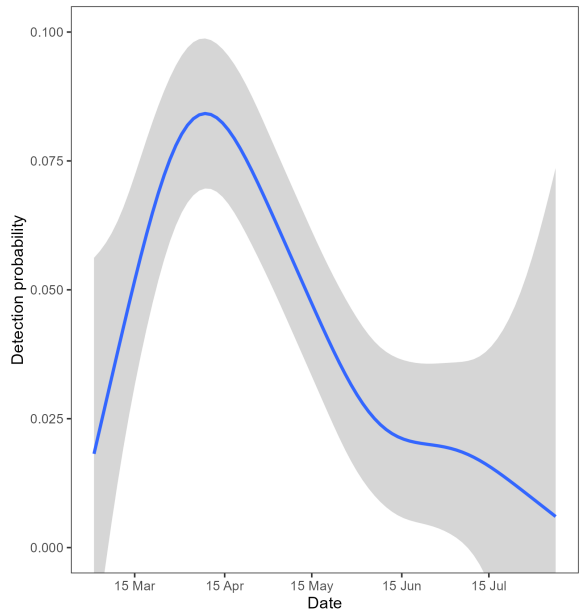
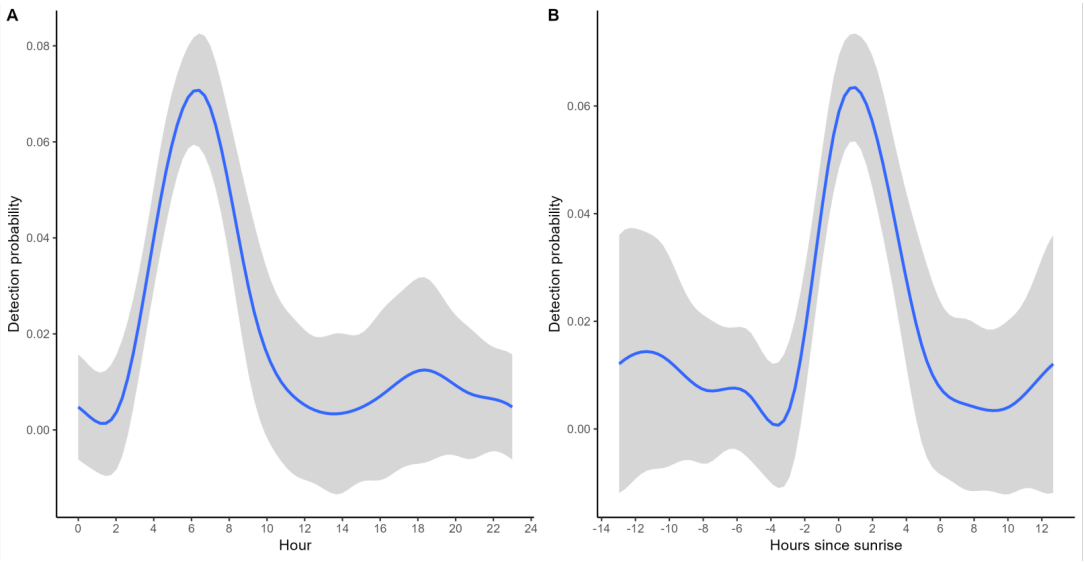


Fig. 4. The predicted probability of detecting a Pileated Woodpecker drum based on a diel scale (hour of the day) (A) and hours since sunrise (B). Predictions generated from a generalized additive model are described in Table 2. The shaded area represents the standard error.



detection window of 2 April, we calculated that a minimum of 22 one-min surveys were required for 90% confidence, and 151 one-min surveys were required for 99% confidence of Pileated Woodpecker presence/absence at a site. For the peak detection window of 06:00 AM, we calculated a minimum of 25 surveys for 90% confidence and 264 surveys for 99% confidence of Pileated Woodpecker presence/absence at a site. By combining peak detection dates (+/− 5 days) and hours (+/− 2 hours), we obtained a detection probability of ~20%. At this detection probability, we calculated a minimum of 10 one-min surveys at 90% confidence and 20 one-min surveys are required to ensure 99% confidence Pileated Woodpecker presence/absence at a site. Approximately 32 one-min surveys are required during typical breeding songbird survey periods (near sunrise in June) to obtain a 90% confidence, or 334 one-min surveys at 99% confidence.

All models that contained environmental variables were significant and explained more variation in the data than the null model (Table 5). The models that explained the most variation in seasonal Pileated Woodpecker drum detection were the mean observed temperature (dynamic variable) and Julian date (static variable) (Table 5). The model, which contained the static variable day length, also explained a comparable amount of variation to the Julian date model (< 1 AIC score difference; Table 5). Therefore, both the Julian date and day length variables were used to test the static and dynamic variable hypotheses (Table 6). The snow cover model explained the least variation (Table 5). Peak drum detection occurred at approximately 0 °C (Fig. 5).

The multivariate GAMM containing day length and mean temperature explained the most variation via AIC scores (Table 6). Both multivariable models outperformed single variable GAMMs described in Table 5. All variables within the multivariable models were described nonlinearly (effective degrees of freedom > 1; Table 6).

Table 5. Model results and Akaike information criterion (AIC) scores for models describing the relationship between Pileated Woodpecker drumming and seasonal predictor variables. Models were produced through a generalized additive mixed effect model, including location as a random effect.

Seasonal predictor model	P-value	Effective degrees of freedom	Degrees of freedom	AIC
NULL	N/A	18.82	19.81	1061.26
Julian date	< 0.005	4.19	24.74	1024.55
Day length	< 0.005	3.21	26.15	1025.01
Mean temperature	< 0.005	4.06	24.42	1018.62
Normal expected mean temperature	< 0.005	3.68	22.49	1030.29
Snow cover	< 0.005	1.11	24.531	1036.73
Days from green up	< 0.005	2.61	22.285	1036.55

Table 6. Model results for multivariate GAMMs that include the most predictive static (Julian date and day length) and dynamic (mean temperature) interannual environmental variables.

Model	Variable	Effective degrees of freedom	P-value	AIC
Julian date + mean temperature	Julian date	2.96	0.015	1012.92
	Mean temperature	3.76	0.037	
Day length + mean temperature	Day length	2.61	0.014	1010.06
	Mean temperature	2.83	0.003	

Results from the generalized linear mixed effects model with location as a random effect failed to find a significant relationship with daily precipitation (p = 0.650) on Pileated Woodpecker drumming (Table 7). However, we did identify a significant

relationship of the deviation from the normal expected mean temperature (p = 0.021; Table 7). The model containing daily precipitation did not explain more variation than the null model. However, the model containing the deviation from normal expected temperature did explain more variation than the null model. The beta coefficient for the deviation from the normal expected mean temperature model demonstrated a positive relationship with Pileated Woodpecker drumming (Table 7).

DISCUSSION

Our results not only provide baseline information for Pileated Woodpecker ecology, in terms of behavioral timing and environmental influences on behavior, but also provide guidelines for optimizing detection in Pileated Woodpecker surveys. We demonstrated how best to use existing non-optimized surveys to obtain high confidence in the accuracy of Pileated Woodpecker habitat use.

The observed pattern of seasonal drumming variation of the Pileated Woodpecker in Alberta supports the idea that Pileated Woodpeckers use drumming to attract mates (Kilham 1959). Similar relationships are supported by Tremain et al. (2008), who demonstrated a seasonal relationship between Floridian Pileated Woodpeckers' drumming and calling behaviors with drumming peaks in late March. However, Pileated Woodpecker drumming occurred throughout our sampling period, suggesting that the species may also use drumming for purposes unrelated to mate attraction or breeding.

We found that the peak drumming occurred on 2 April in the boreal forest at the northern edge of the species' range. This peak drumming timing differs between Florida and Alberta, suggesting that latitudinal variation and potentially different cues may trigger changes in Pileated Woodpecker drumming behavior. These results provide key information to optimize survey timing for future Pileated Woodpecker surveys in northern environments. However, it is important to note that lower temperatures and inclement weather in many parts of the world that can be common in March/April may affect ARU battery life and reduce the observer's ability to properly detect Pileated Woodpeckers. How often poor weather influences the ability to detect Pileated Woodpeckers in March/April should be compared to May/June. Furthermore, we did not demonstrate latitudinal effects in our study. This could be due to our study's relatively small extent compared to the North American range of the Pileated Woodpecker. It is likely that data from across the species' range would be required to observe this latitudinal effect on Pileated Woodpecker drumming behavior.

Our findings suggest that breeding songbird surveys that are concentrated in June do not adequately capture the peak detection windows for Pileated Woodpeckers. During these standard breeding songbird survey timings (near sunrise in June), a minimum of 32 surveys should be conducted to obtain high confidence about Pileated Woodpecker use of a site. Furthermore, whether drumming at this time of the year reflects evidence of breeding activity becomes less certain. More work is needed to determine if seasonal changes in drumming behavior show differences in the vegetation types where drumming is observed. This would help us understand where breeding signals are given relative to nest versus foraging locations. This survey recommendation underscores the value of passive acoustic monitoring. Given the time and financial expenses associated with repeated visits for human observation point counts, passive acoustic monitoring generally offers a more practical solution. With this approach, researchers can efficiently conduct the recommended 32 surveys by making just two trips to the location: once for deployment and once for retrieval. However, human observation point counts are typically longer than the one-min recording length used in this study; this could in theory reduce the number of surveys required to achieve a high confidence. Sampling many one-min recordings instead of a few longer recordings allows for the increased likelihood of the detection of a rare event/vocalization because of temporal autocorrelation. This approach is particularly important for species that vocalize infrequently.

Fig. 5. The relationship between mean temperature and Pileated Woodpecker drumming detection probability as estimated by the smoothing term generated in the GAMM. The shaded area represents the bounds of the 95% confidence intervals.

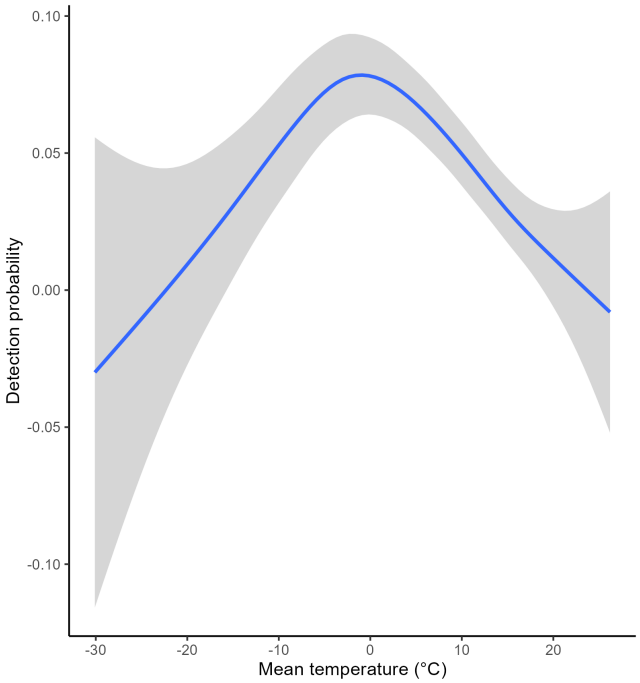


Table 7. Results of the generalized mixed effects models describing the relationship between Pileated Woodpecker drumming behavior and daily fluctuations in weather described by the model variable.

Model variable	Beta coefficient	Standard error	P-value	AIC
NULL	N/A	N/A	N/A	1130.32
Precipitation	−0.013	0.030	0.650	1132.10
Deviation from the normal expected mean temperature	0.037	0.016	0.021	1127.03

We found a relationship between Pileated Woodpecker drumming and time of day. We observed that Pileated Woodpeckers were more likely detected by drumming during the morning, around sunrise. Most songbird species follow a similar daily pattern to what we observed with Pileated Woodpecker drums (Kacelnik and Krebs 1983, Staicer et al. 1996, Bruni et al. 2014), suggesting that the woodpecker drum is analogous to birdsong. Additionally, we observed a secondary peak in drumming activity near dusk; many avian species demonstrate a similar pattern of crepuscular activity (Catchpole and Slater 1995, Staicer et al. 1996, Hannah et al. 2022). A possible explanation for this change in drumming activity is that the sunlight, visibility, and/or temperature in midday hours may be more conducive for activities such as foraging or nest maintenance.

The optimal daily timing of Pileated Woodpecker drums occurs simultaneously with typical songbird singing patterns (Kacelnik and Krebs 1983, Staicer et al. 1996, Bruni et al. 2014). Therefore, the daily timing of breeding songbird surveys maximizes the likelihood of detecting a Pileated Woodpecker drum. We recommend that surveys of Pileated Woodpeckers in Alberta, Canada (or locales within similar latitudinal/environmental boundaries) begin in the first week of April to ensure the greatest likelihood of detection. It is most beneficial to focus sampling near sunrise to increase the likelihood of detecting Pileated Woodpeckers. Specifically, one hour from sunrise is the optimal time for detecting Pileated Woodpecker drums. Combining the seasonal and daily optima requires at least ten 1-min recordings to ensure 90% confidence of Pileated Woodpecker presence/absence. A caveat of this recommendation is that the Pileated Woodpecker is a highly mobile bird with an expansive home range often larger than a point count sampling radius (Renken and Wiggers 1989, Mellen et al. 1992, Bull and Holthausen 1993, Bonar 2001, Yip et al. 2017). Thus, it may not be using a particular location on a given day. Therefore, distributing these recordings across multiple days will increase the probability of detecting the species as they move throughout their home range. Transect surveys or ARU clusters that cover the entirety of a Pileated Woodpecker's home range could be employed to increase probability of detecting this species. Additionally, the spatial clustering of surveys/ARUs may increase the likelihood of population closure within a sampling unit, which is a crucial assumption required by many occupancy and abundance models. Although further research is needed to determine the correct spatial scale of an ecologically meaningful sampling unit for this species, we emphasize that these survey recommendations are a minimum requirement to achieve desired confidence. Because we calculated our detection probability only from sites where we observed Pileated Woodpeckers, we established an improved index for Pileated Woodpecker survey requirements based on locations where presence was already known.

The single-variable model of mean temperature explained the most variation in the seasonal detection of Pileated Woodpecker drums, suggesting that the initiation of the Pileated Woodpecker breeding season is linked to conditions that may change yearly. This finding suggests that this species may be flexible to changes in phenology (Stenseth et al. 2002, Visser et al. 2012, Samplonius et al. 2018). However, as we further investigated this relationship, we discovered that static variables such as Julian date and day length, in addition to mean temperature, better explain variation in seasonal Pileated Woodpecker drumming behavior. This suggests that Pileated Woodpeckers are both hard-wired to begin breeding behaviors (increased frequency of drumming) based on a combination of the local environmental conditions and the static patterns of seasonal change. These relationships are nonlinear in nature, suggesting that Pileated Woodpeckers prefer conditions that are not too late in the season and not too hot or cold temperature for drumming. We observed that the peak detection of Pileated Woodpecker drums occurred when the daily mean temperature was approximately 0 °C, and detection was highest in April. It is probable that the early onset of warmer conditions (above freezing) may be linked to Pileated Woodpecker drumming behavior. Furthermore, these findings may be useful to researchers outside of our study area to determine when Pileated Woodpeckers in their area may be more active (and therefore detectable).

Dynamic environmental variables, such as the observed temperature, are the most predictive variables in predicting Pileated Woodpecker drumming frequency, which may suggest that this species' drumming behavior is flexible on the basis of the present environmental conditions. This behavioral plasticity may be important as climatic conditions become increasingly more unpredictable (Dawson 2008, Noonan et al. 2018). Food availability is a limiting factor in reproductive success and survival for all birds (Martin 1987, Dawson 2008). The timing of peak invertebrate egg hatching, and therefore prey abundance, is often linked to temperature (Powell and Logan 2005, Dunn et al. 2011, Nadolski et al. 2021). Thus, warmer spring temperatures could create a phenological mismatch between invertebrate phenology and avian nest initiation. The combination of static and dynamic interannual variables that influence Pileated Woodpecker drumming suggests that this species may be less susceptible to phenological mismatches created by climatic change than migratory species. However, additional research should examine how Pileated Woodpeckers react if the optimal conditions are not met. Additionally, limited research has been conducted on the seasonality of the Pileated Woodpecker's primary food source, carpenter ants (McClelland and McClelland 1999, Powell and Logan 2005, Bull and Jackson 2020). We suggest further research to investigate the environmental triggers influencing carpenter ant egg hatch timing and whether these factors correlate with drumming frequency, nest initiation, probability of nest occupancy, and hatching or fledging success. It would also be useful to assess whether drumming activity is predictive of nest occupancy and/or breeding status for Pileated Woodpeckers. Additionally, it is important to note that all of the environmental variables explored in this analysis (mean temperature, normal expected mean temperature, snow cover, and days from green-up) proved to influence the Pileated Woodpecker drumming. Therefore, it is likely that multiple static and dynamic interannual variables could influence Pileated Woodpecker behavior.

We found a relationship between the deviation from the expected normal temperature and Pileated Woodpecker drums, suggesting that daily weather can influence Pileated Woodpecker drumming behavior. This result further supports our findings that temperature influences the seasonal/interannual variation in drumming. The observed positive relationship suggests that Pileated Woodpeckers are more likely to drum on warmer-than-average days. Furthermore, we found no relationship with

precipitation. However, weather is often variable across a landscape. Weather stations used in this study were occasionally > 10 km from the site, and even stations as close as 1 km from the survey site may not accurately represent local weather conditions.

Prior to this study, there had been no formal attempt to standardize acoustic sampling for Pileated Woodpeckers in Alberta. Current survey standards employ data collection during the songbird breeding season, but our results demonstrate that Pileated Woodpecker drumming activity and detection are sub-optimal during this period. Modeling attempts using this sub-optimal period may only partially capture how the species uses the landscape, especially for breeding purposes. We recommend that future modeling and survey efforts for Pileated Woodpeckers in Alberta be focused during the optimal periods of March/April. From our calculated detection probabilities, land managers can reliably assess whether potentially suitable Pileated Woodpecker habitat occurs within their operating areas. This is increasingly important for land managers because of regulations put in place by the Government of Canada's Section 70 of the Migratory Birds Regulations (2022), stating that Pileated Woodpecker nesting cavities may not be disturbed for 36 mo after being deemed unoccupied. This regulation affects industries such as the forestry and energy sectors, which are prevalent in Alberta and have competing interests with the Pileated Woodpecker's associated habitats of old-growth forests (Bull and Holthausen 1993, McClelland and McClelland 1999, Aubry and Raley 2002, Hartwig et al. 2004, Krementz et al. 2012). Following this study's temporal survey guidelines and minimum survey effort recommendations, these industries can more accurately detect Pileated Woodpecker occupancy in operating areas and conduct informed nest searches within areas of known occupancy. Further research should be conducted to determine the relationship between ARU-derived site occupancy and Pileated Woodpecker nest occupancy.

Author Contributions:

A. C. Zeller was responsible for concept formation, data analysis, and writing the manuscript. E. M. Bayne was responsible for providing data, concept formation, and writing the manuscript. C. L. Mahon provided feedback on concept formation and writing of the manuscript.

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Data Availability:

Annotated code for analyses can be found online in the open repository: https://github.com/austinczeller/Temporal_PIWO.

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APPENDIX

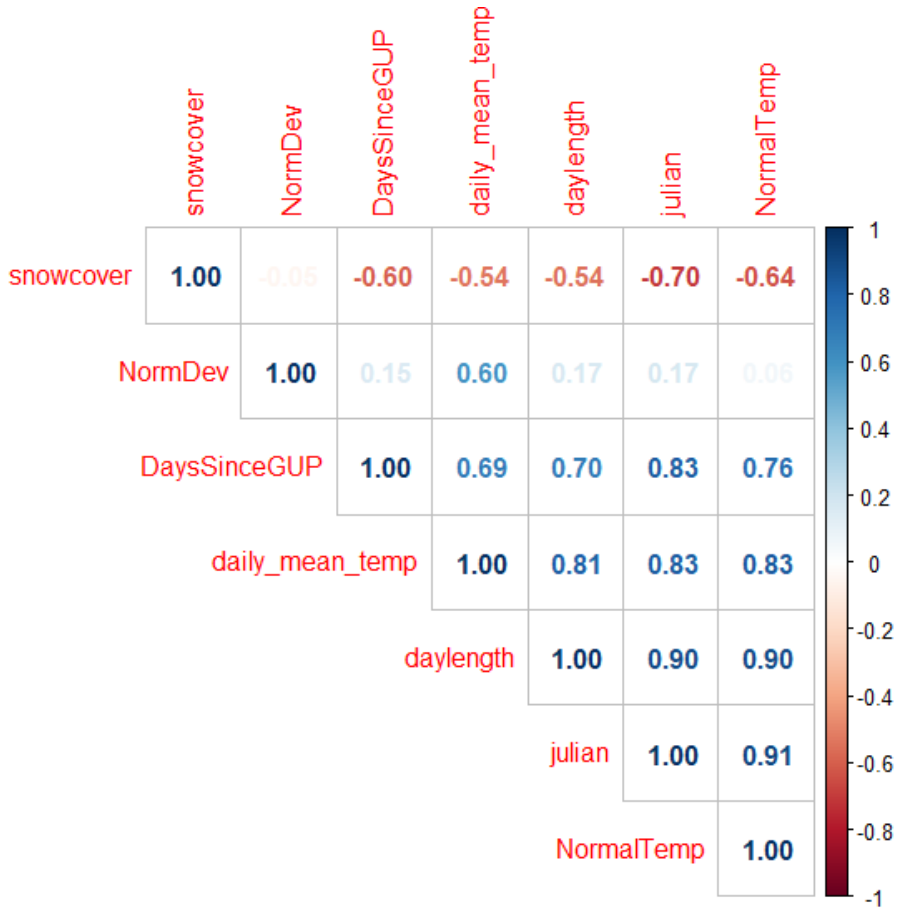

Figure 1.A Correlation matrix for variables examined in objective 2: environmental factors influencing Pileated Woodpecker drumming. "NormDev" is the deviation from the expected normal temperature and "DaysSinceGUP" is the days since green up.

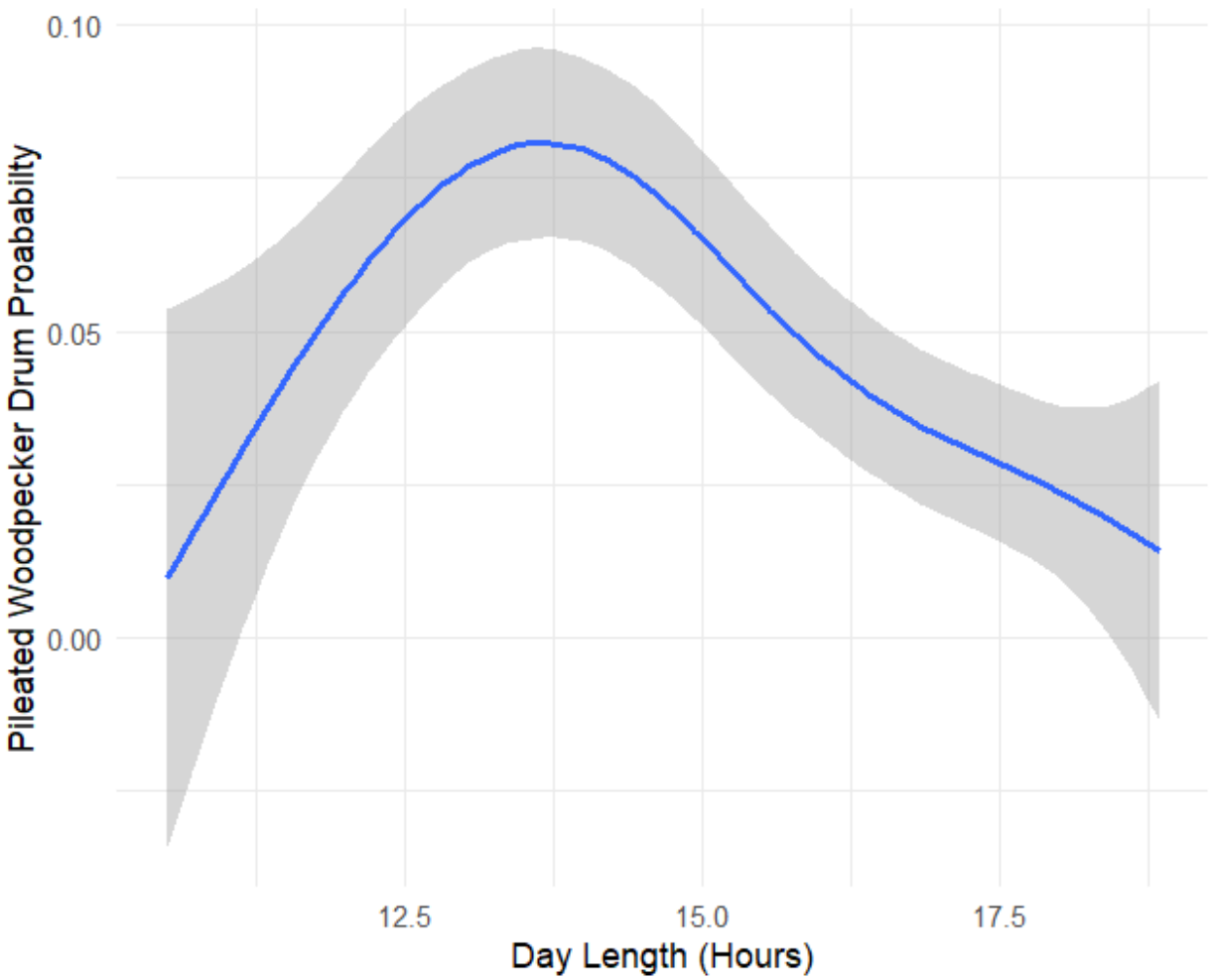

Figure 2.A Relationship of day length (hours) and Pileated Woodpecker drumming as estimated by the smoothing term generated via GAMM. The shaded area represents the bounds of the 95% confidence intervals.

Table 1.A Model results for multivariable models with/without interactions between variables explored in Chapter 2, objective 3. Interaction term represents the interaction between precipitation and deviation from the expected mean temperature.

Model	Variable	Beta coefficient	P-value	AIC
Precipitation + deviation from the expected mean temperature	Precipitation	-0.009	0.755	1128.929
	Deviation from the expected mean temperature	0.038	0.022	
Precipitation X deviation from the expected mean temperature	Precipitation	-0.033	0.376	1128.026
	Deviation from the expected mean temperature	0.043	0.009	
	Interaction Term	-0.013	0.075	

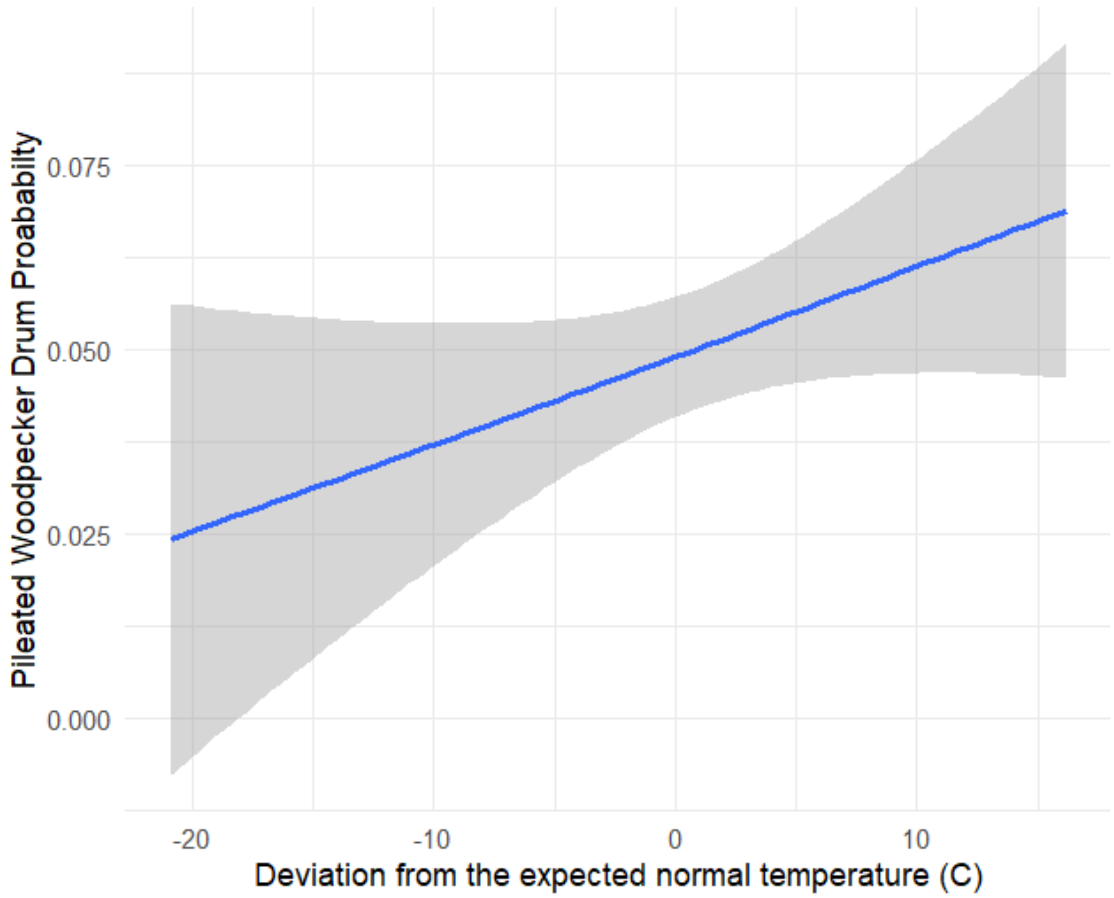

Figure 3.A The relationship between the deviation from the expected normal temperature (Celsius) and the probability of detecting a Pileated Woodpecker drum. Shaded area represents the 95% confidence interval.